

Symbolic regression: Reconstruction of knowledge

Related papers

GoodRegressor: [Jang](#), arXiv:2510.1832 → Oxygen ion conductors, NASICON, superconducting oxides (under revision)

Oxygen ion conductors: [Jang et al.](#) Sci. Data **13**, 778 (2026).

Hydrogen storage materials (metal alloy): [Jang et al.](#), Chem. Sci. **16**, 23111 (2025). **selected as a “hot paper”**
[Jang et al.](#), Chem. Sci. (2026). DOI: 10.1039/D6SC03089K

Social Psychology: Yu & [Jang et al.](#), PsyArXiv. 10.31234/osf.io/waqjn_v1

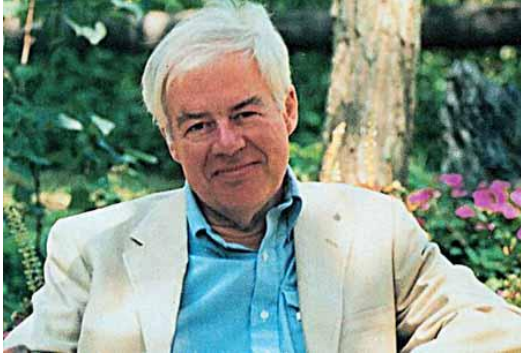
FYI,

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South Korean, Bachelor (2010) & master (2012) in Tohoku Univ., PhD (2020) in uTokyo.

Postdoc in Kyoto Univ. & NIMS, Assistant prof. in IMR & AIMR.

Self-introduction



Richard Rorty (1931-2007)

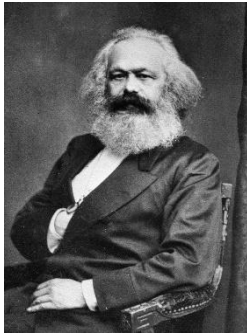
When an American philosopher, Richard Rorty, was asked about his personal life during an interview, he replied, if I remember correctly, “When I was young, I was **not prepared for small talk.**”
Me neither.

Instead of discussing where I am from, let me share my interests.

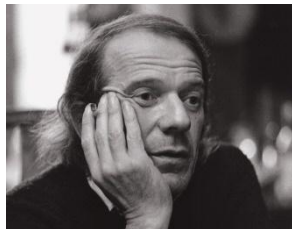
Philosophy



Georg Hegel
(1770-1831)



Karl Marx
(1818-1883)

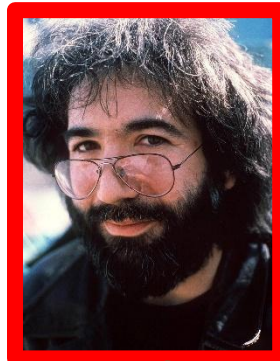


Gilles Deleuze
(1925-1995)



Terence McKenna
(1946-2000)

Music



Jerry Garcia
(1942-1995)



Lou Reed
(1942-2013)



Wataru Takada
(1949-2005)



Kiyoshiro Imawano
(1951-2009)

Comedy



Ryutaro Kamioka
(1942-2023)



John Sullivan
(1946-2011)



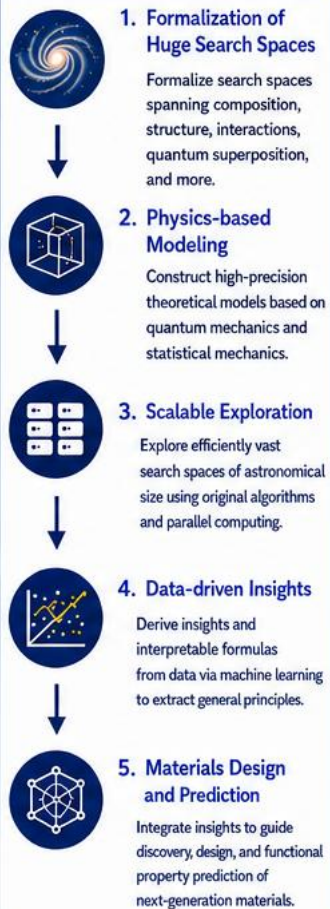
Andy Kaufman
(1949-1984)



Bill Hicks
(1961-1994)

From quantum to materials macroscale: uncovering astronomical search spaces with scalable computation

Research Flow

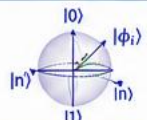


Astronomical Search Spaces



Quantum Superposition

$$|\Psi\rangle = \sum_i c_i |\phi_i\rangle$$



Composition



Combinations of element types and their ratios ...

Structure



Various atomic arrangements, defects, interfaces, etc. ...

Interactions

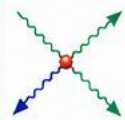


Various interactions such as bonding, long-range, local, spin, lattice, and quantum correlations ...

Overall Picture of Our Research

Large-scale Theoretical Computing
× Original Algorithms
High Accuracy · High Speed · Scalable

Quantum Perturbation Calculation (SQPerturbation)



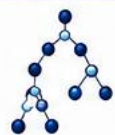
Evaluate high-order quantum correlations efficiently and achieve high accuracy with many-body effects.

Solid Optimization on Lattices (EwaldSolidSolution)



Treat long-range interactions rigorously and evaluate astronomically large-scale problems with high accuracy using Ewald-type algorithms.

Symbolic Regression (GoodRegressor)



$$y = a_1 x_1^2 + a_2 x_1 x_2 + \sin(a_3 x_3) + \dots$$

Discover interpretable formulas from data to unveil complex, nonlinear relationships.

High-Performance Computing Infrastructure (HPC)

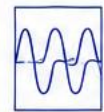


Optimize massively parallel computations at scales of tens of thousands to millions of cores to maximize memory efficiency and communication.

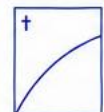
High-accuracy Prediction and Discovery



Stable structures and phase transition prediction



Electronic structure, properties, and dynamics prediction



Nonlinear response and functional property prediction



Mechanism elucidation and design principles



Construction of large-scale data assets and reusable knowledge

Data Science Applications



Material Discovery & Screening
Efficiently explore high-dimensional design spaces to identify promising candidates.



Acceleration of Materials Design
Predict properties before experiments and drastically shorten development cycles.



Understanding Complex Phenomena
Clarify principles behind complex phenomena such as strong correlation and disorder.



Breaking the Scale Barrier
Overcome the limits of conventional algorithms and solve previously intractable large-scale problems.

Software Development & Provision



SQPerturbation
Quantum perturbation calculation package



EwaldSolidSolution
Solid optimization on lattices package



GoodRegressor
Symbolic regression package

We provide these packages as open, scalable, and user-friendly software for the research community.

Key Strengths & Expertise



Deep Theoretical Expertise
Expertise in quantum many-body theory, statistical mechanics, and exploration of huge search spaces.



Original Algorithm Development
Design, implement, and optimize novel algorithms tailored for large-scale problems.



Large-scale Parallel Computing
Leverage HPC to the fullest for high-performance programming and scalable implementations.



Data Analysis & Modeling
Extract laws from large-scale data via machine learning and modeling; ensure interpretability and rigorous validation.



Open & Collaborative Software Development
Develop software openly and promote reproducible and extensible research.

Through original computational methods and data analysis, we unlock the potential of materials and contribute to a sustainable and prosperous future.

Symbolic? Towards a new scientific paradigm

Conventional Paradigm I: Simplicity as Truth

- Scientific truth should be expressible by simple laws and equations.
- Successful examples from classical science reinforced this belief.
- However, not all natural phenomena are necessarily simple.

Conventional Paradigm II: Prediction Without Understanding

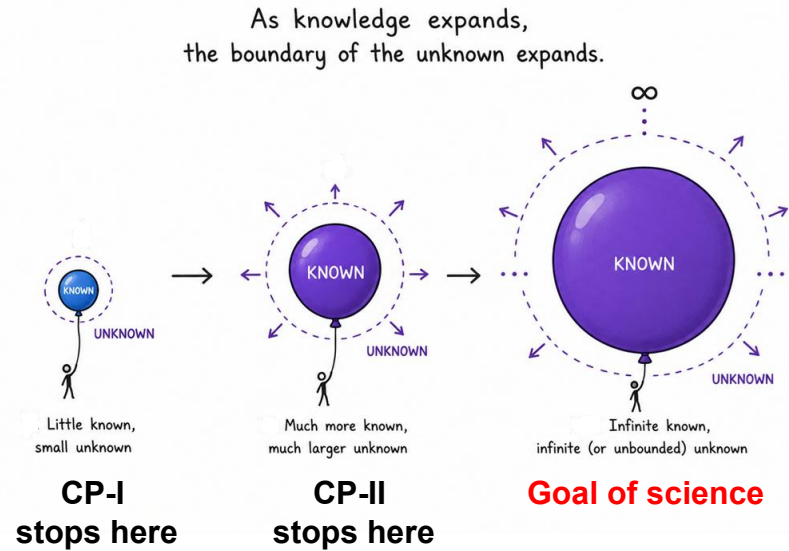
- Complex phenomena are assumed to require complex templates.
- Predictive performance is often prioritized over interpretability.
- Understanding the internal structure is frequently regarded as optional.

A Proposed Paradigm

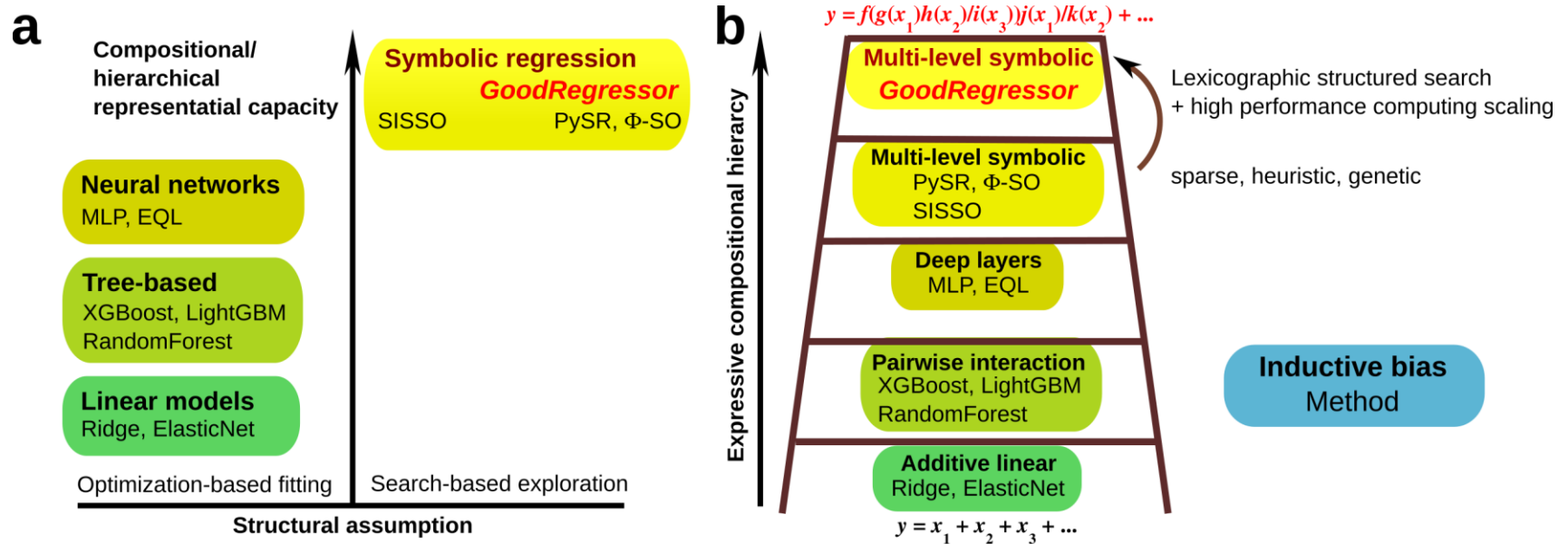
- Nature may contain both simple and intrinsically complex phenomena.
- The degree of complexity should itself be quantified.
- Scientific progress should not stop at prediction or template construction.
- Even when a phenomenon requires a highly complex representation, science should strive to reveal and interpret every constituent element.
- If current methodologies are insufficient for such understanding, developing new methodologies becomes a scientific objective in its own right.

Vision

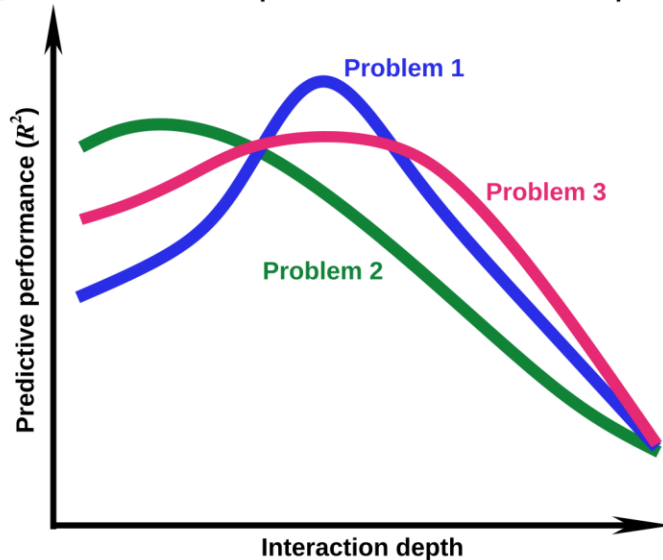
- Neither simplicity for its own sake nor black-box prediction is the ultimate goal.
- The goal of science is a quantitatively grounded and human-interpretable understanding of reality, regardless of its complexity.
- Machine learning, symbolic regression, first-principles modeling, and future methods are merely tools toward that objective.



Inductive bias of *GoodRegressor*



c Non-monotonic performance vs. interaction depth



$y = x_1 + x_2 + x_3 + \dots$

$y = f(g(x_1)h(x_2)/i(x_3))j(x_1)/k(x_2) + \dots$

Characteristics of the optimal interaction depth

Problem 1: Narrow optimum at high interaction depth

An explicit solution likely exists,
but it is difficult to identify using conventional tools.

Problem 2: Broad optimum at low interaction depth

An explicit solution exists and
can be readily identified using standard tools.

Problem 3: Broad optimum at high interaction depth

The solution itself is intrinsically complex and
remains difficult to discover using conventional tools,
but can be effectively captured with *GoodRegressor*..

Algorithm grammar of GoodRegressor

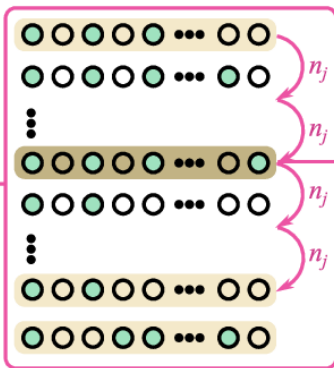
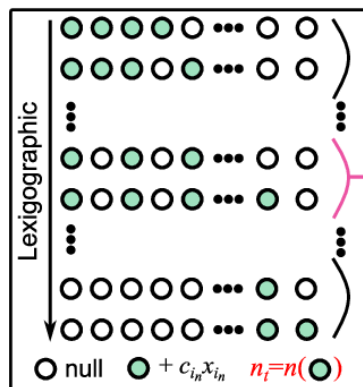
a run through (the jungle)

$$\mathbf{X}_1 = \{x_1, x_2, \dots, x_{i_c}\} \cup \{x_{i_c} x_{i_j}, \dots\} \cup \{x_{i_c} / x_{i_j}, \dots\}$$

$$\mathbf{X}_1^o, \mathbf{X}_1^m, \mathbf{X}_1^d: i_p, i_j = 1, \dots, i_c$$

$$\mathbf{X}_1^d: \text{sign}[\min(x_{i_j})] \text{sign}[\max(x_{i_j})] = 1$$

$$= \{x_1, x_2, x_3, x_4, x_5, \dots, x_{i_c+1}, x_{i_c+2}\}$$

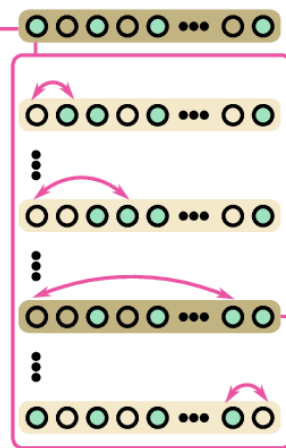


n_j is adjusted to meet the target calculation time.

Model M_{i_c} : highest R^2_{vald} under F_f for

F_f : "full Fisher" condition = $p(F)$ and $\forall x_{i_n}, p(t) < 0.05$
or Frequentist

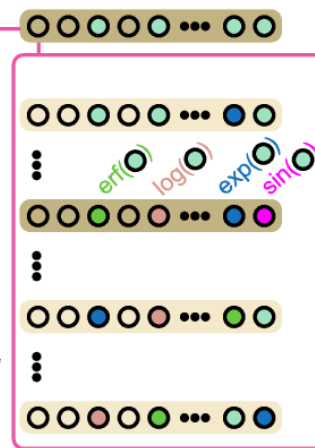
b swap



Highest R^2_{vald} under F_f

Update M_{i_c} if $R^2_{\text{vald}} \uparrow$

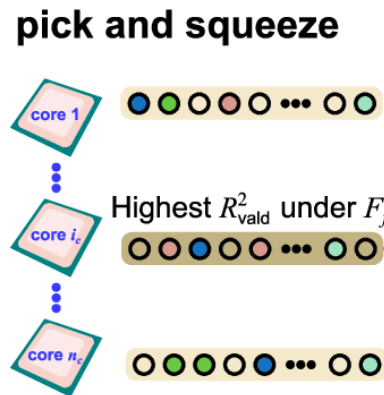
c transit



Highest R^2_{vald} under F_f

Update M_{i_c} if $R^2_{\text{vald}} \uparrow$

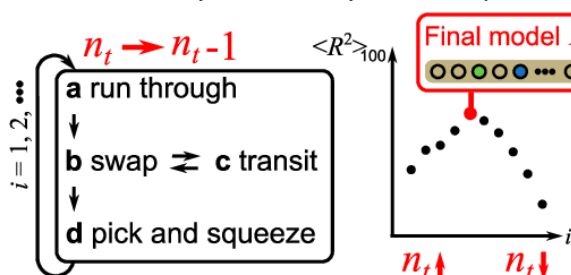
d pick and squeeze



$$\mathbf{X}_i = \{x_1, x_2, \dots, x_{i_c}\} \cup \{x_{i_c} x_{i_j}, \dots\} \cup \{x_{i_c} / x_{i_j}, \dots\}$$

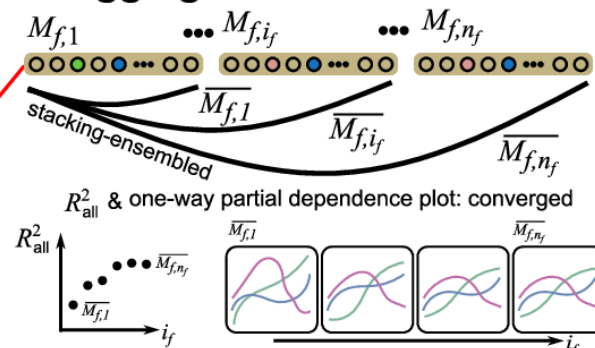
$$\mathbf{X}_i^o, \mathbf{X}_i^m, \mathbf{X}_i^d: i_p, i_j = 1, \dots, i_c$$

$$\mathbf{X}_i^d: \text{sign}[\min(x_{i_j})] \text{sign}[\max(x_{i_j})] = 1$$



e bagging

varying train-test divisions



Robustness $\uparrow$ Noise-fitting and risk of overfitting $\downarrow$

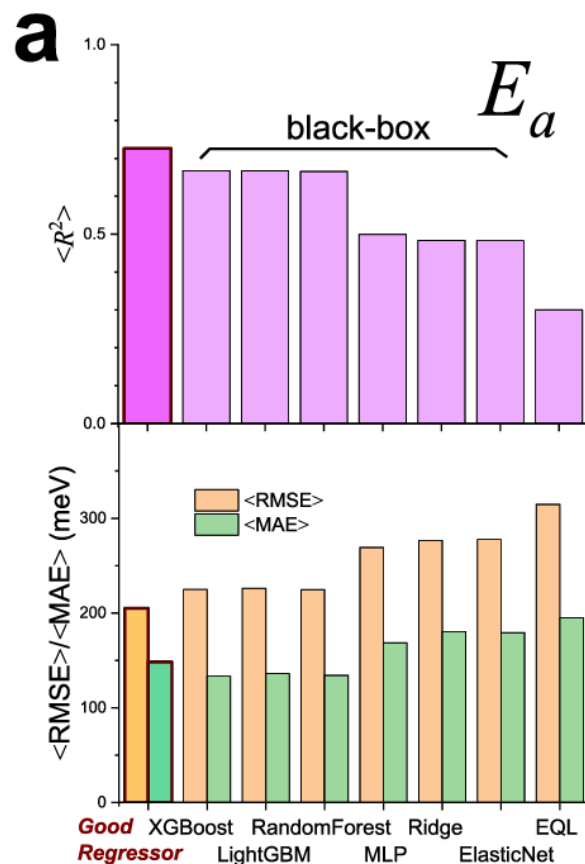
Codes: <https://github.com/JerryGarcia1995/OxygenIonConductor>

MPI-C++, all was made from scratch (except for using a linear algebra library "Eigen").

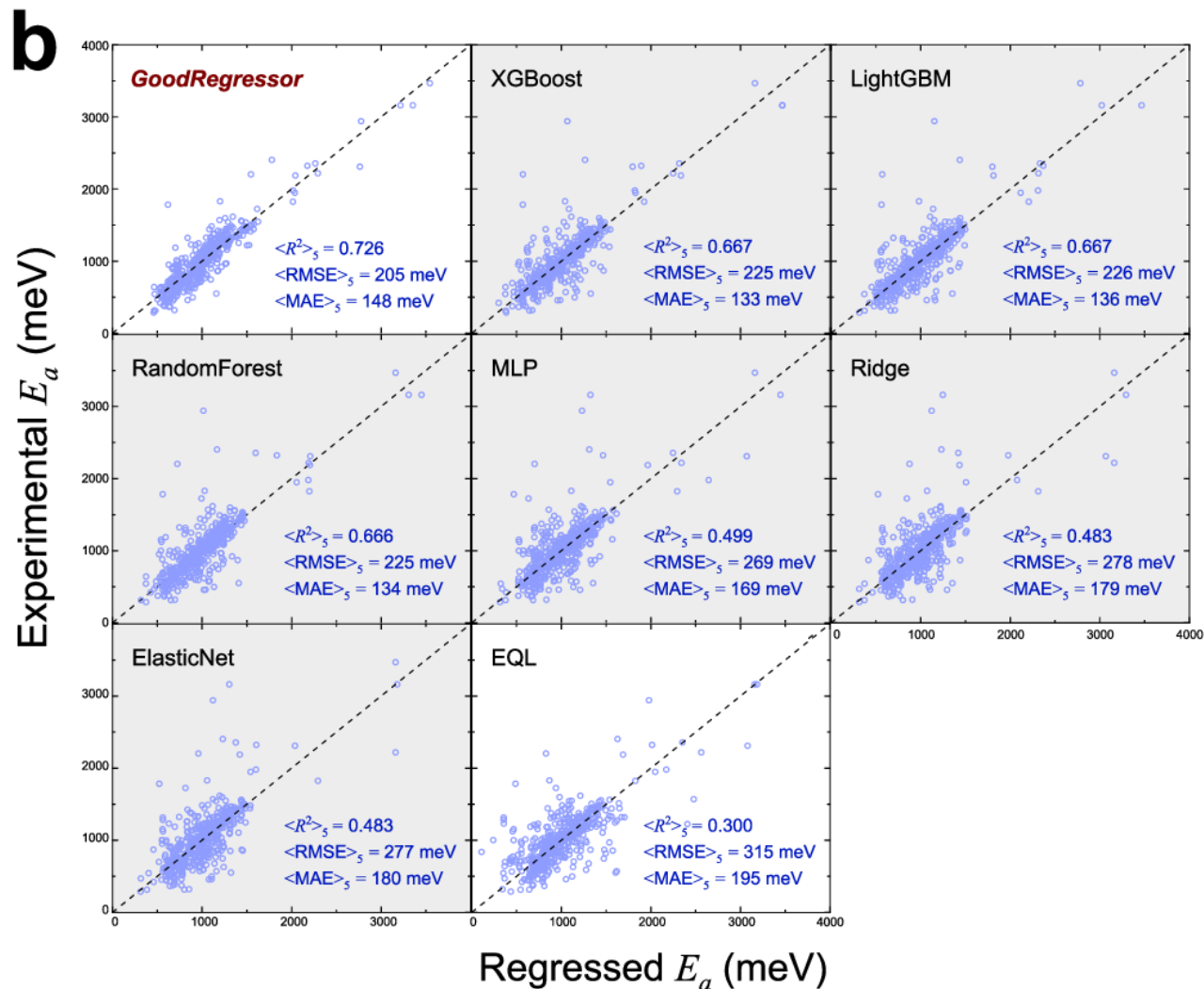
GoodRegressor beats black boxes

Oxygen ion conductors: activation energy

of candidate equations: 1.44×10^{457}



Φ -SO, PySR: $\langle R^2 \rangle < 0$
 SISSO: memory issue

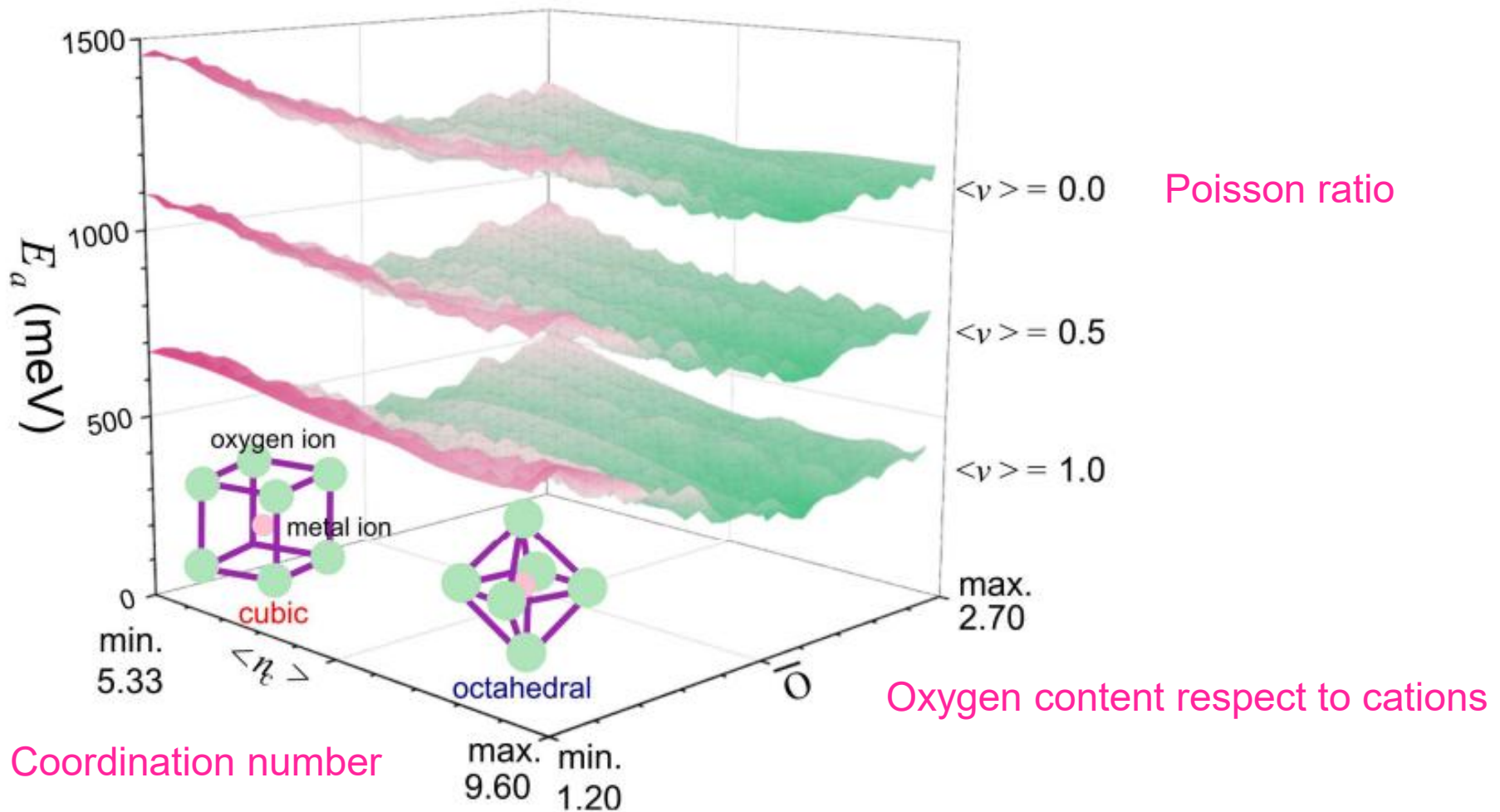


Promising?

the apatite-type compound $\text{La}_{9.5}\text{Si}_{5.5}\text{Al}_{0.5}\text{O}_{26}$. Jang, arXiv:2510.1832

GoodRegressor beats black boxes

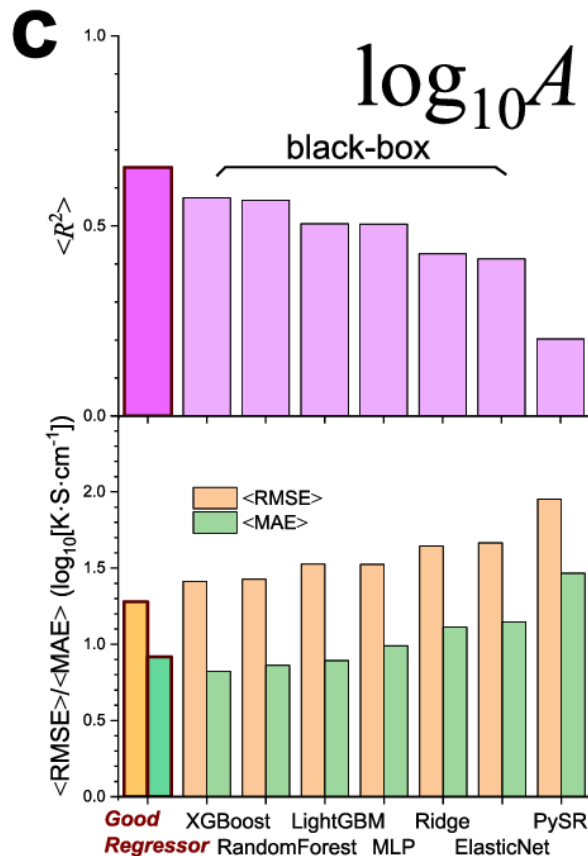
Which “the interactions of terms” are important? Never answered by the state-of-the-art MLs



GoodRegressor beats black boxes

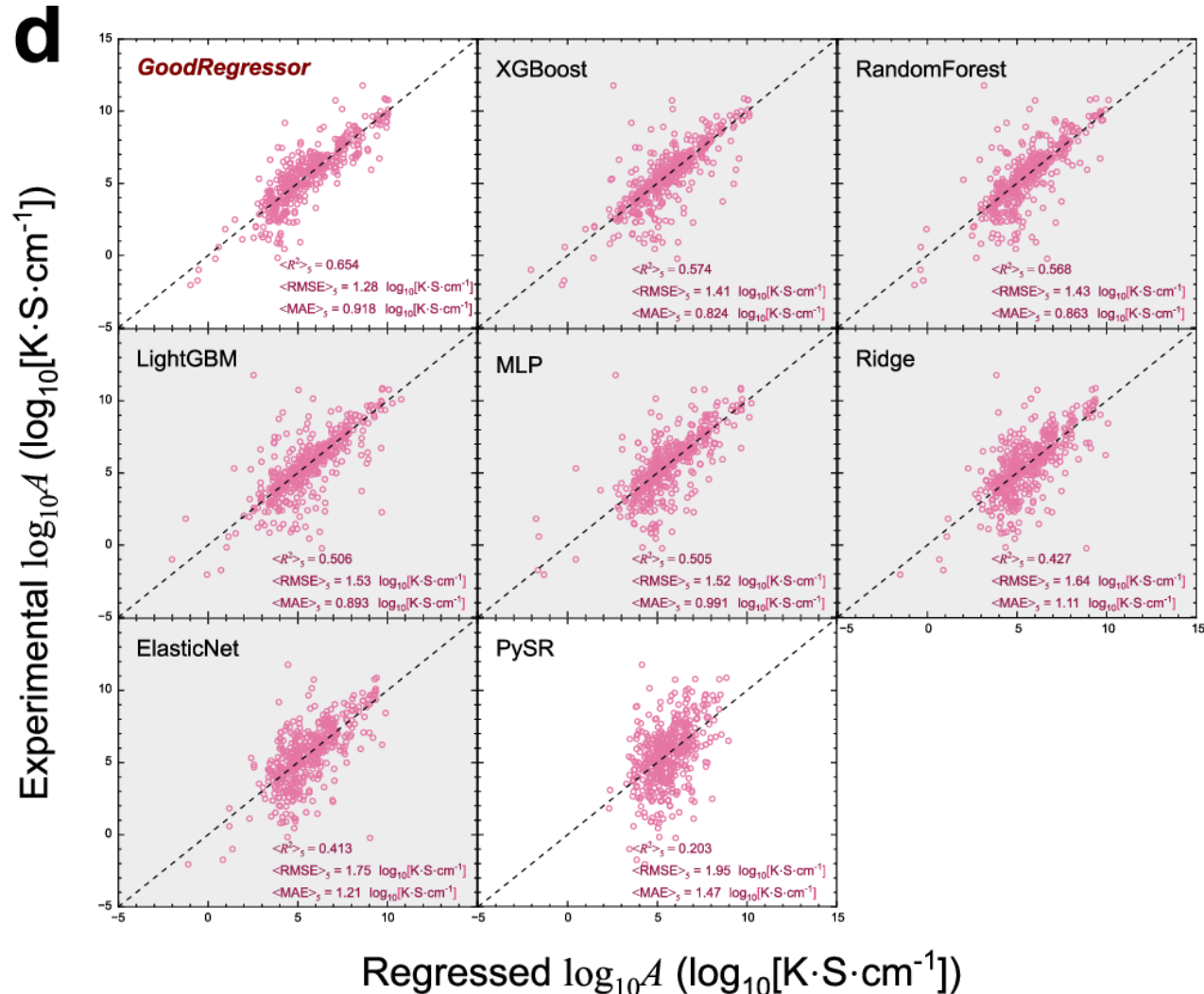
Oxygen ion conductors: Arrhenius prefactor

of candidate equations: 1.44×10^{457}



Φ -SO, EQL: $\langle R^2 \rangle < 0$

SISSO: memory issue



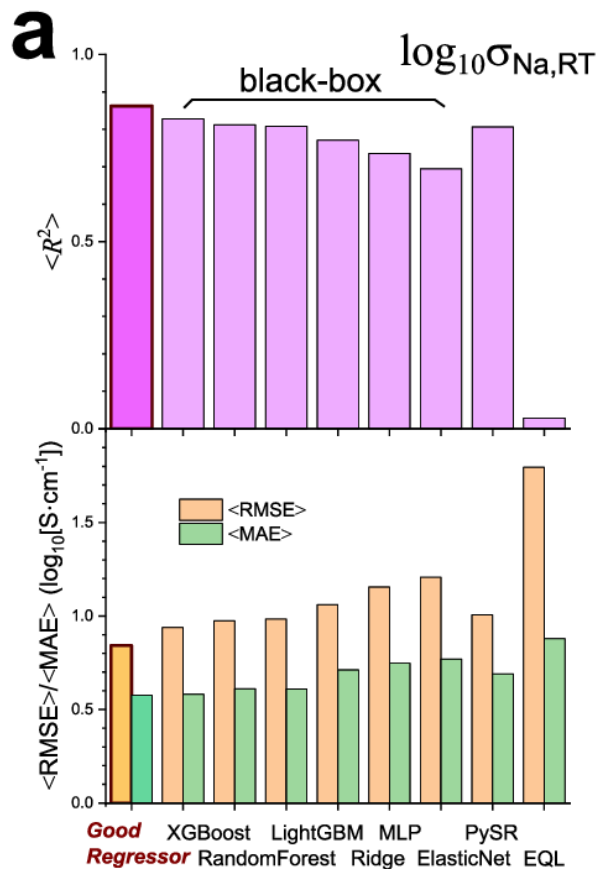
Promising?

the apatite-type compound $\text{La}_{9.5}\text{Si}_{5.5}\text{Al}_{0.5}\text{O}_{26}$. Jang, arXiv:2510.1832

GoodRegressor beats black boxes

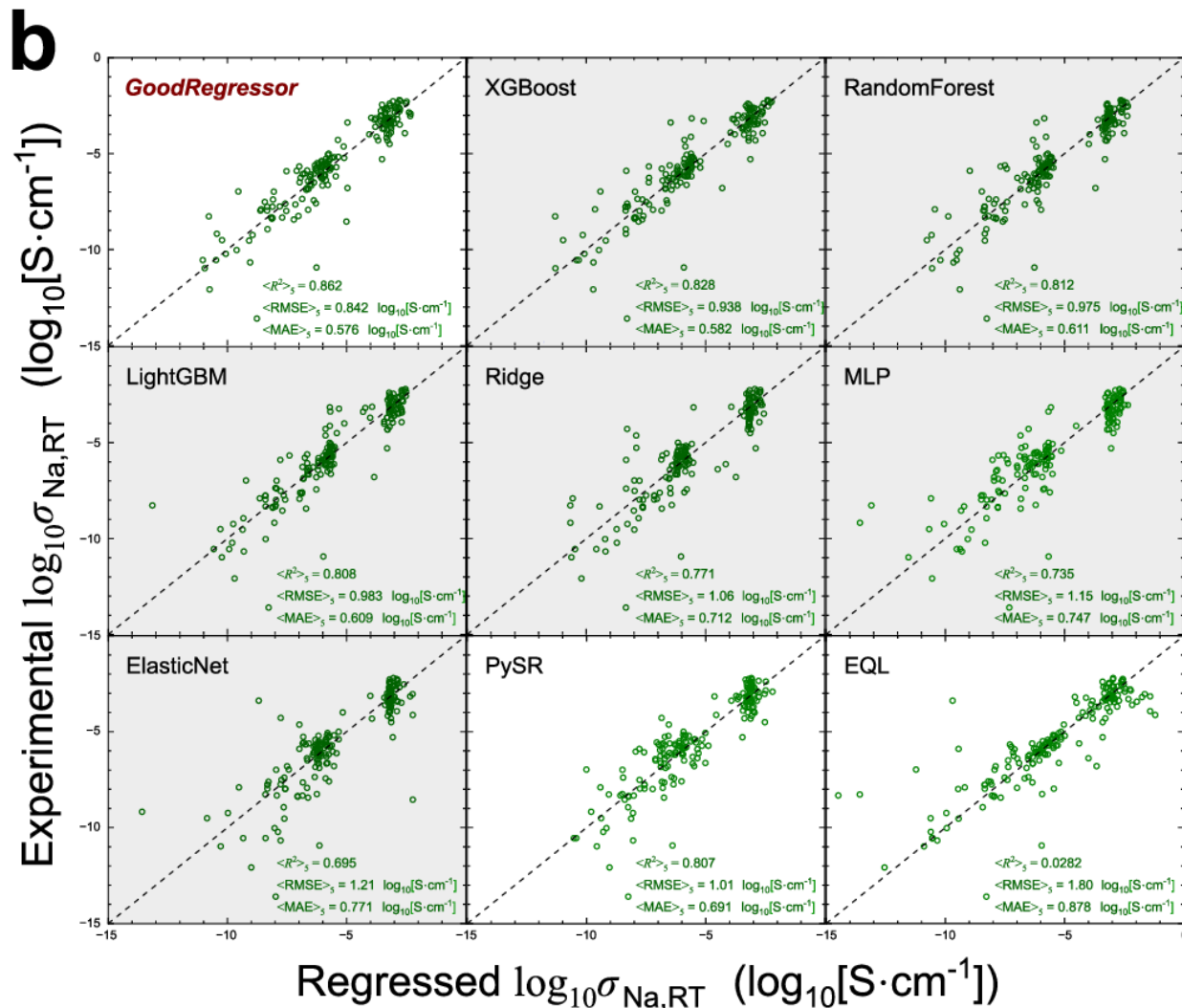
NASICON: Na-ion conductivity at room temperature

of candidate equations: 5.99×10^{124}



$\Phi\text{-SO}$: $\langle R^2 \rangle < 0$

SISSO: memory issue

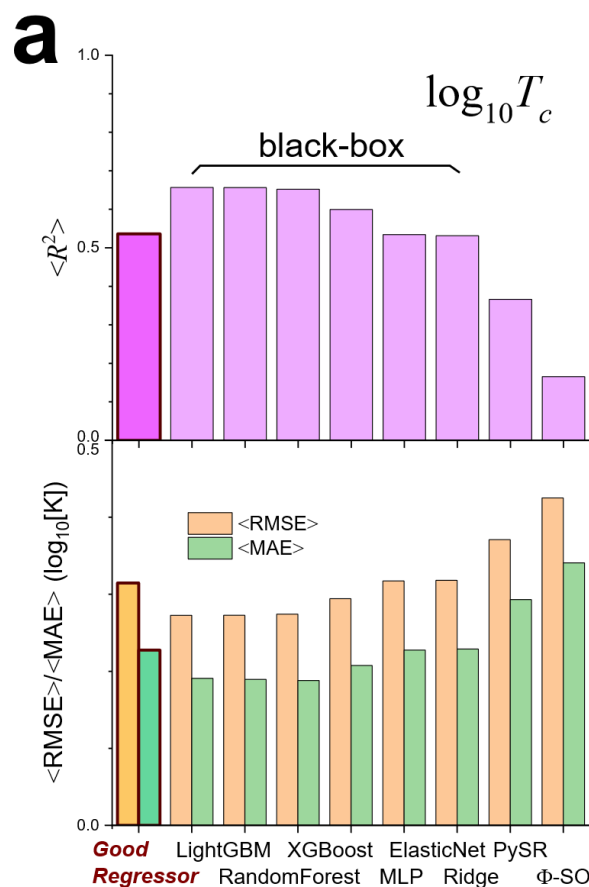


Promising?

A Zr-free composition, $\text{Na}_{3.4}\text{Y}_{0.4}\text{Hf}_{1.6}\text{Si}_2\text{PO}_{12}$ Jang, arXiv:2510.1832

GoodRegressor beats black boxes

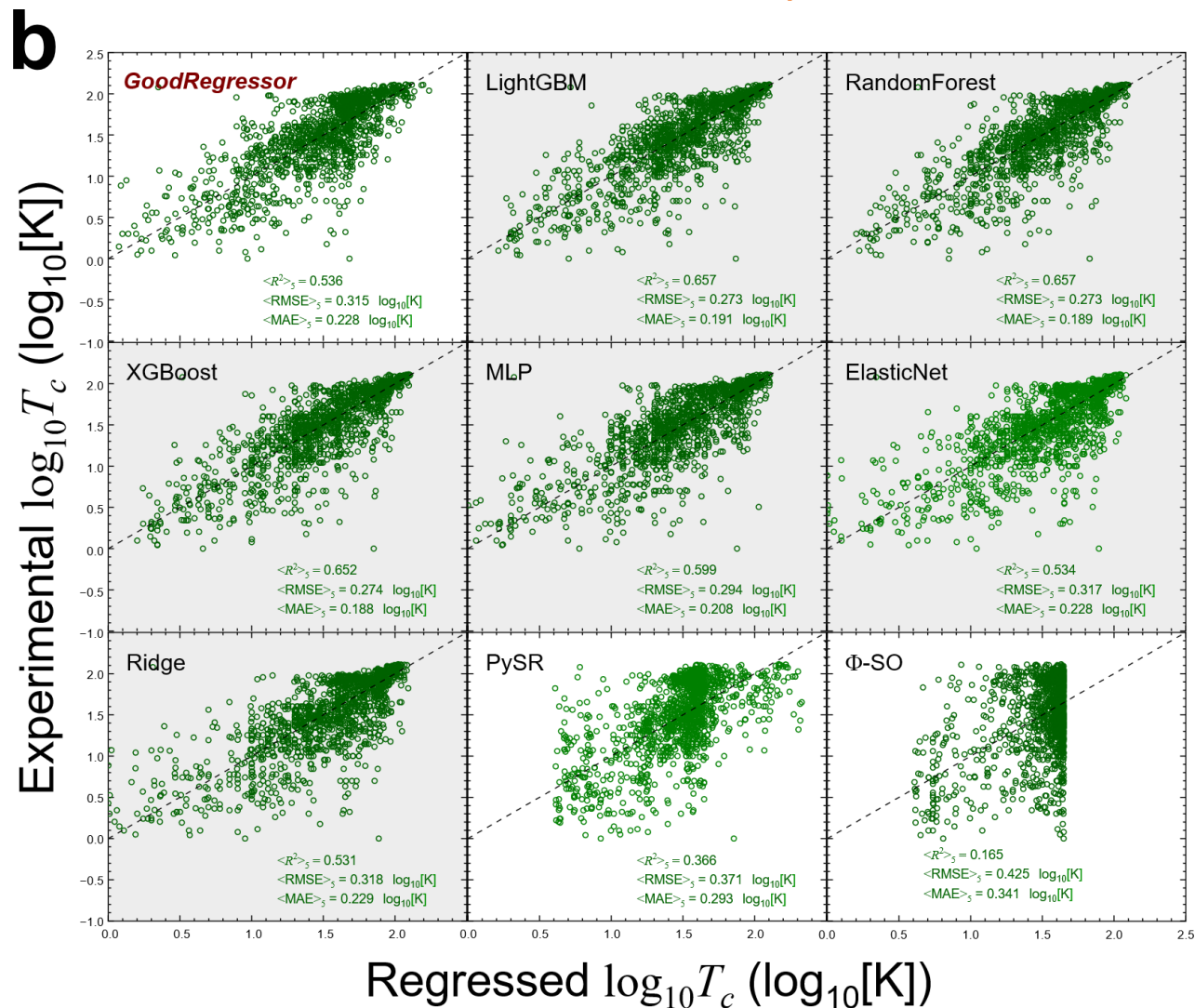
Superconducting oxides: Transition temperature # of candidate equations: 4.20×10^{430}



EQL: $\langle R^2 \rangle < 0$

SISSO: memory issue

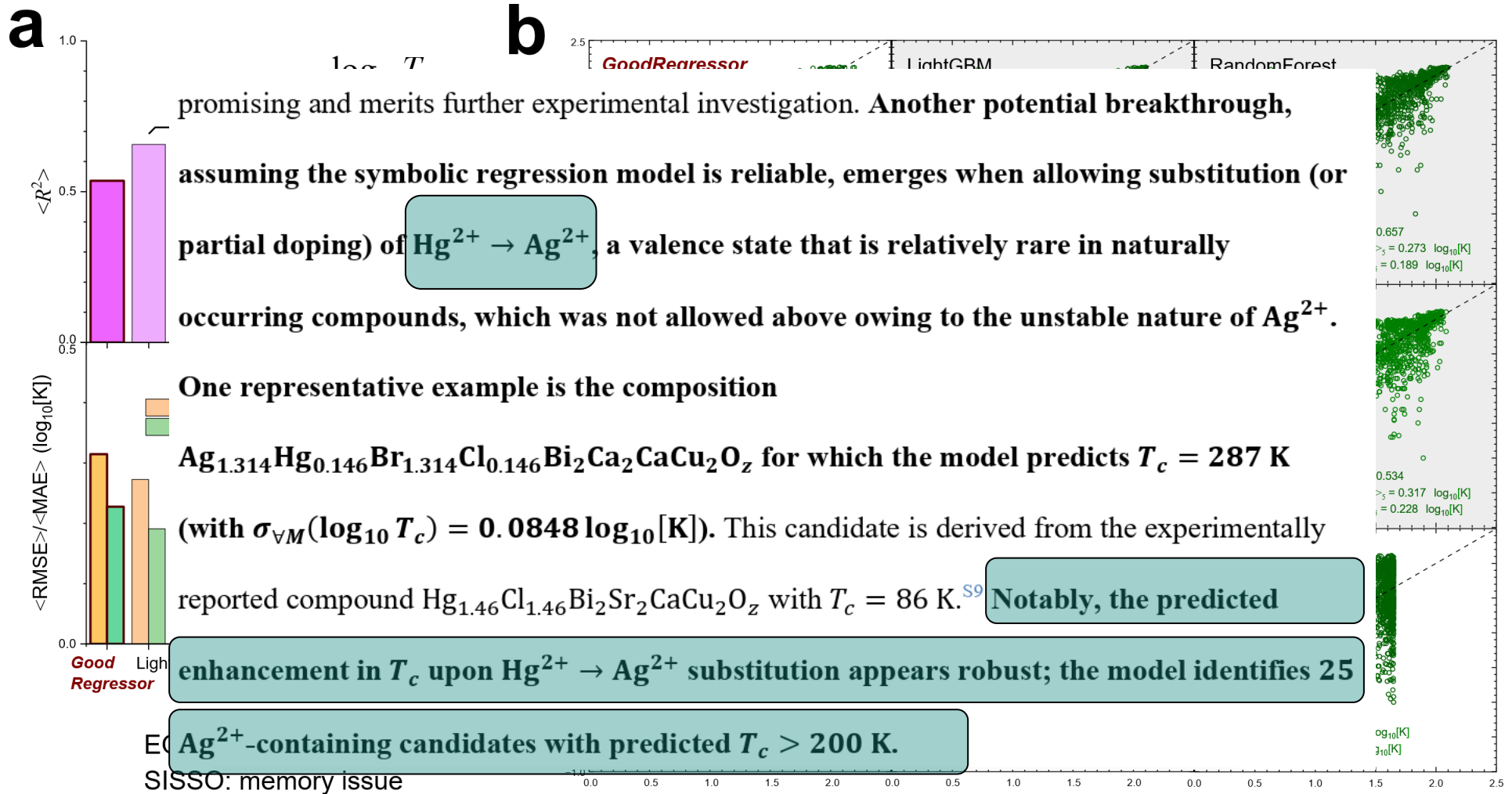
Promising?



Jang, arXiv:2510.1832

GoodRegressor beats black boxes

Superconducting oxides: Transition temperature # of candidate equations: 4.20×10^{430}



Promising?

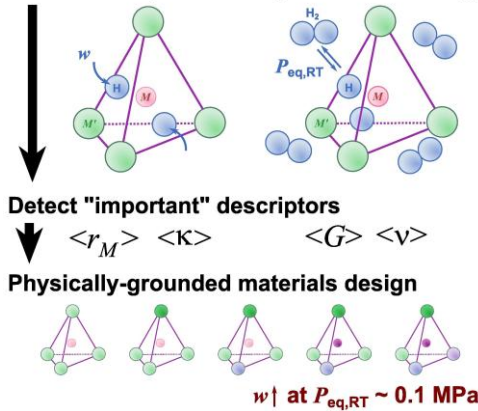


Regressed $\log_{10} T_c$ ($\log_{10}[\text{K}]$)

Jang, arXiv:2510.1832

Hydrogen study: What this talk will answer

White-box "symbolic" regression modeling



Why interstitial hydrides?

Which descriptors govern w and $P_{eq,RT}$?

Materials design

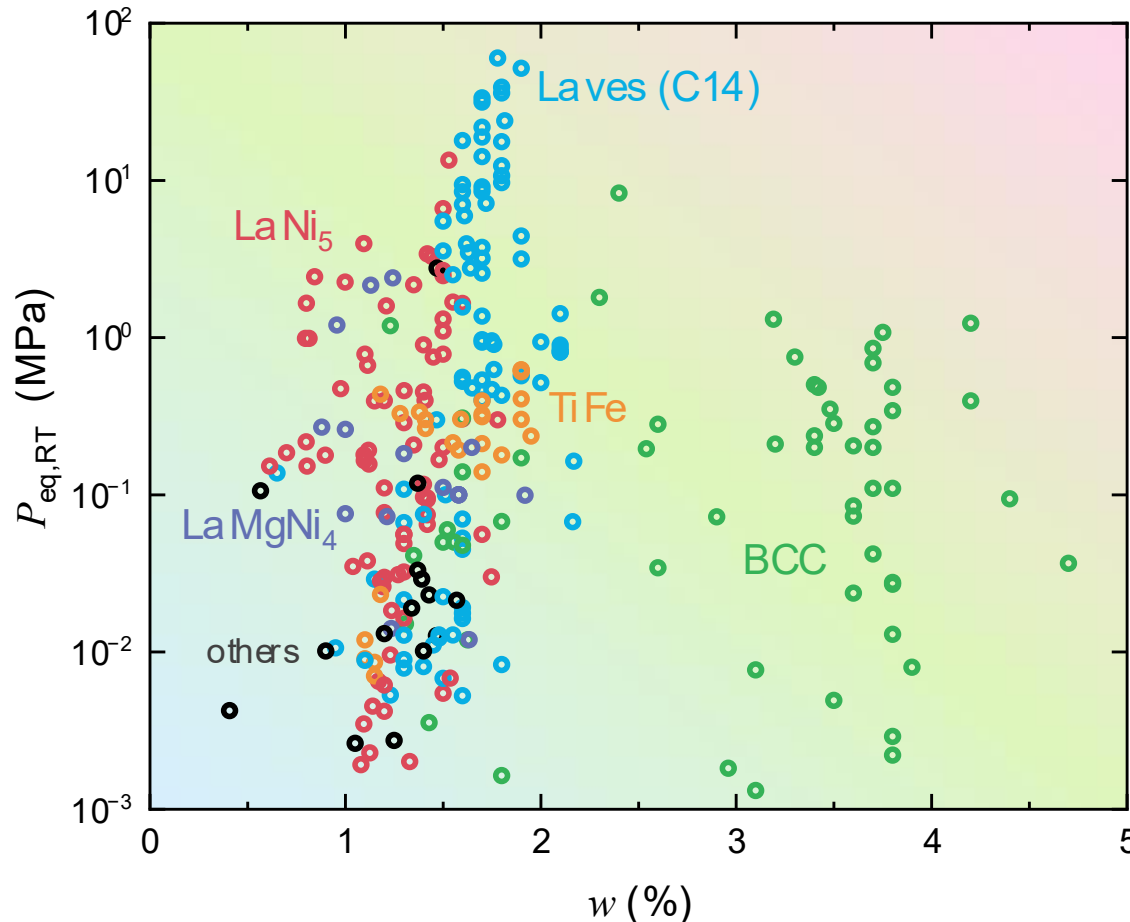
Follow-up studies & perspectives

Central idea: use curated PCT data and white-box symbolic regression to translate descriptor-property relationships into materials design rules.

Target: high hydrogen capacity (w) at practical room-temperature equilibrium pressure ($P_{eq,RT} \sim 0.1 \text{ MPa}$).

Hydrogen storage challenge: capacity vs. usable pressure

DigHyd database (interstitial hydrides only)



- Solid-state hydrides offer compact and reversible H₂ storage.
- Interstitial hydrides have favorable kinetics and practical pressures, but intrinsically limited gravimetric capacity.

• **Design goal: raise w without pushing $P_{eq,RT}$ away from ambient-pressure operation.**

Design window used here:

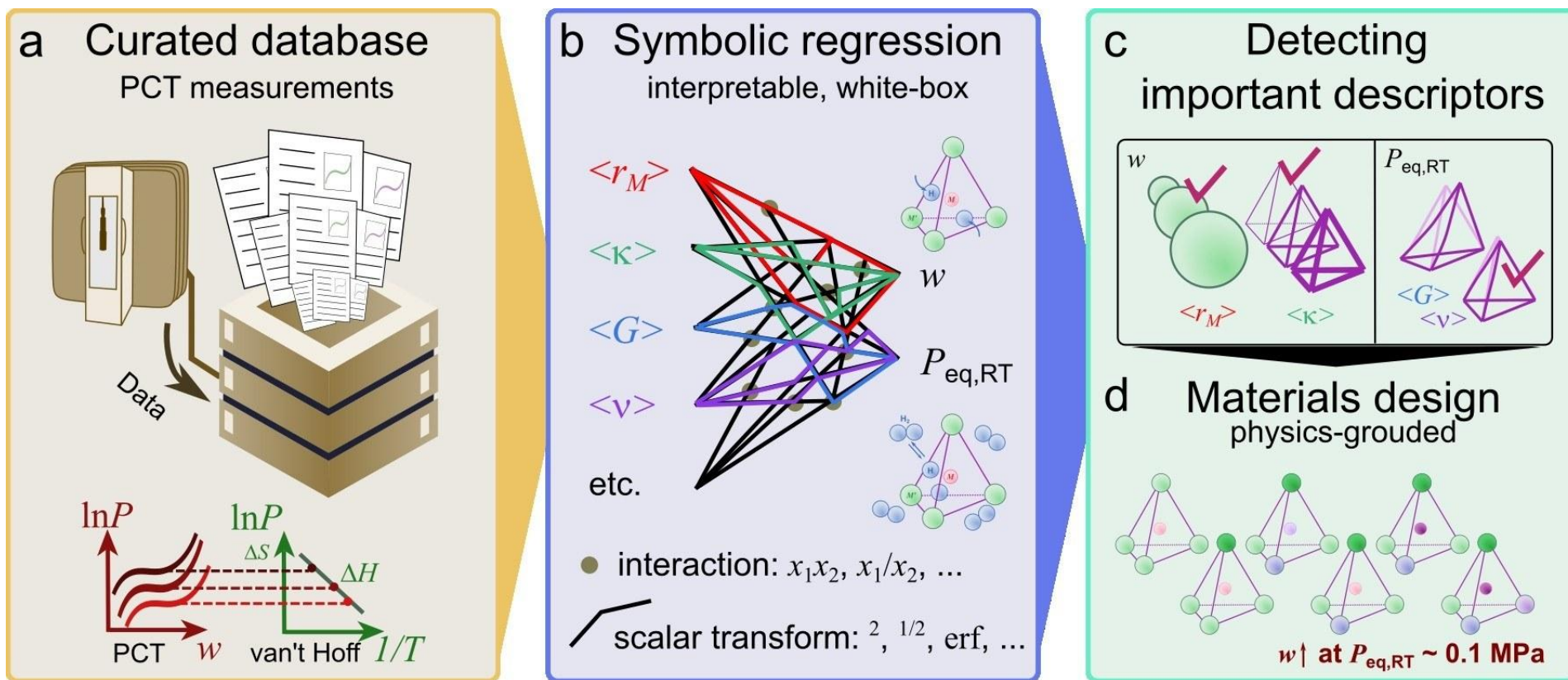
$$\begin{aligned} -1.5 \log_{10}[\text{MPa}] &< \log_{10} P_{eq,RT} \\ &< -0.5 \log_{10}[\text{MPa}] \end{aligned}$$

near practical ambient conditions

Question

Can we find interpretable descriptors that independently explain capacity and equilibrium pressure? Can we perform physics-aware materials design?

Workflow: curated PCT data to physics-grounded design



DigHyd

×

GoodRegressor

- *DigHyd* provides curated experimental PCT measurements.
- *GoodRegressor* builds interpretable symbolic models for w and $P_{eq,RT}$.
- Important descriptors are converted into composition optimization pathways.

For *DigHyd*, Jang et al., Chem. Sci. **16**, 23111 (2025); Zhang et al., Chem. Sci. **17**, 3031 (2026);
Jang et al., arXiv:2603.14139 (2026).

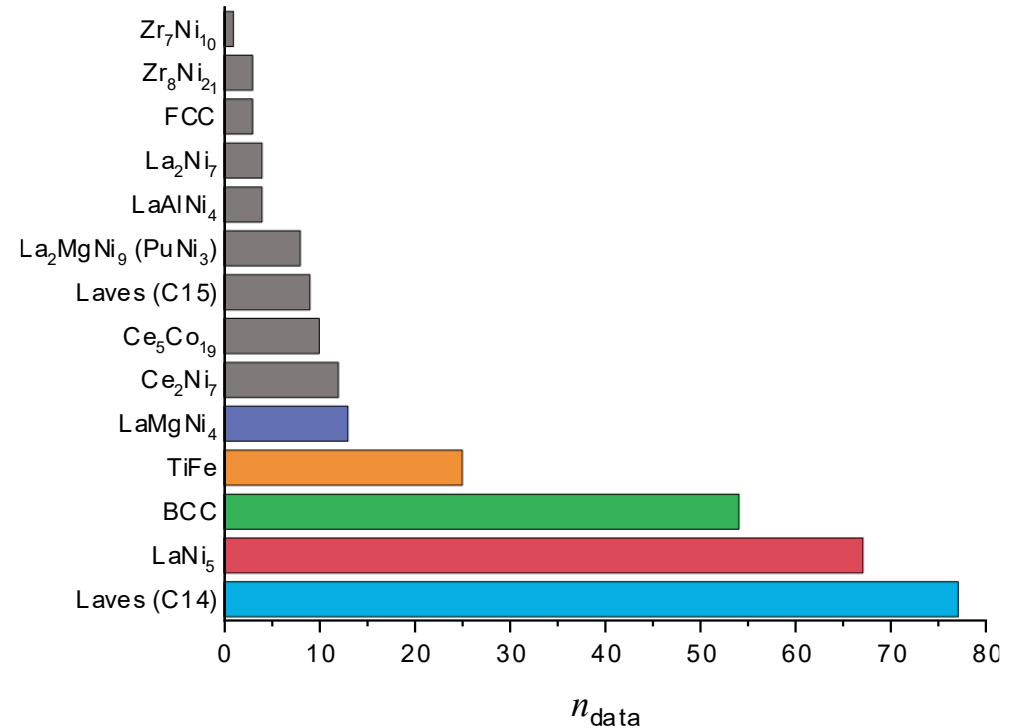
For *GoodRegressor*, Jang, arXiv:2510.1832

Data curation: reduce noise before regression

Notes on data curation

- PCT experiments often include multi-phase materials; these can induce noise-fitting.
- This study retains single-phase or near-single-phase cases (> 80 wt.% dominant phase).
- w model: $n_{\text{data}} = 706$ entries.
- $P_{\text{eq,RT}}$ model: $n_{\text{data}} = 299$ entries with multi-temperature PCT data.

Dominant data classes



Laves (C14), LaNi₅, BCC, and TiFe systems cover much of the multi-temperature PCT space, but the final models are checked for class robustness.

Feature construction: chemical, physical, structural

Chemical

electronegativity
atomic mass
valence electron
count

Physical

metal radius, density
Bulk modulus, shear
modulus, Poisson
ratio
Thermal conductivity,
thermal expansion
coefficient, Debye
temperature, Molar
magnetic
susceptibility

Structural

coordination number
solid angle
solid-angle deviation
polyhedron volume

Compositional

transition-metal
fraction
4th-row transition
metal fraction
transition + rare-earth
metals fraction

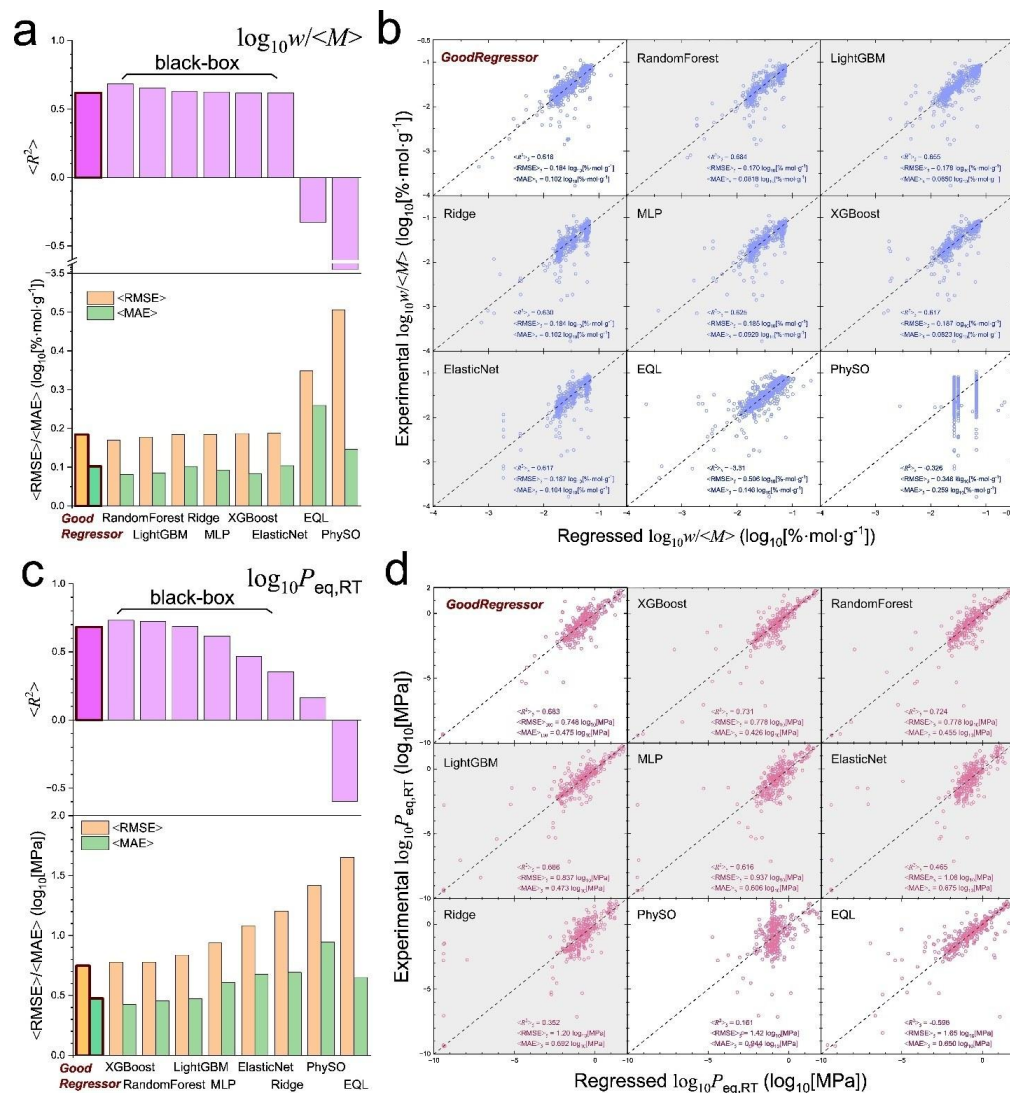
Candidate descriptors are generated from each composition and, when relevant, the parent structure.

**For each elemental descriptor: average $\langle \dots \rangle$, standard deviation $\sigma(\dots)$,
and skewness $r(\dots)$ are constructed
over constituent metal ions.**

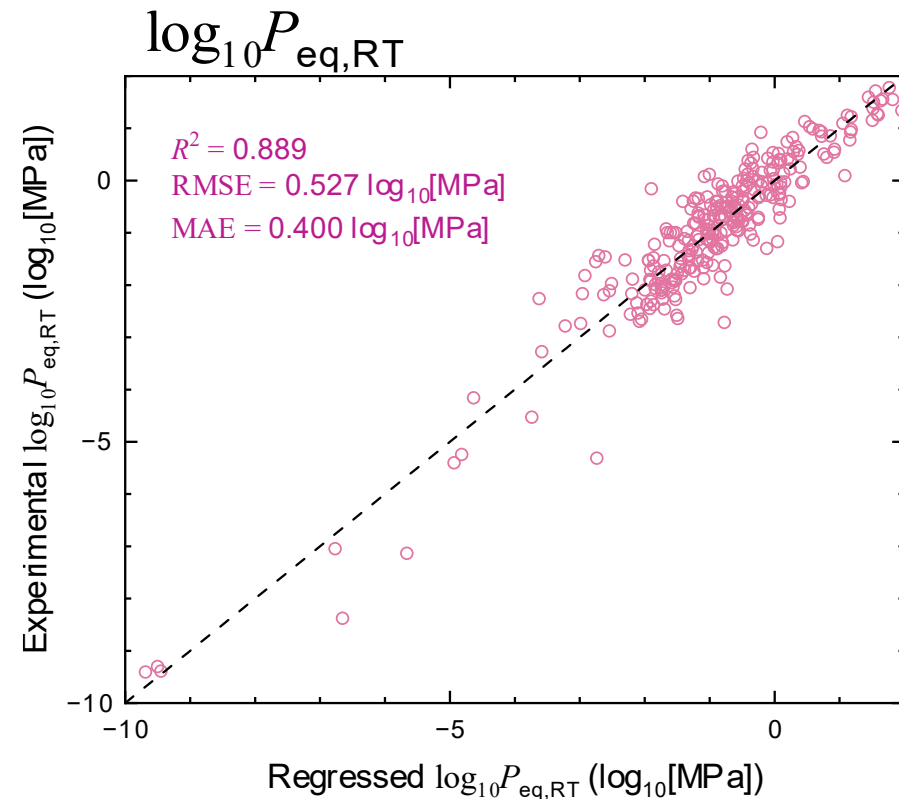
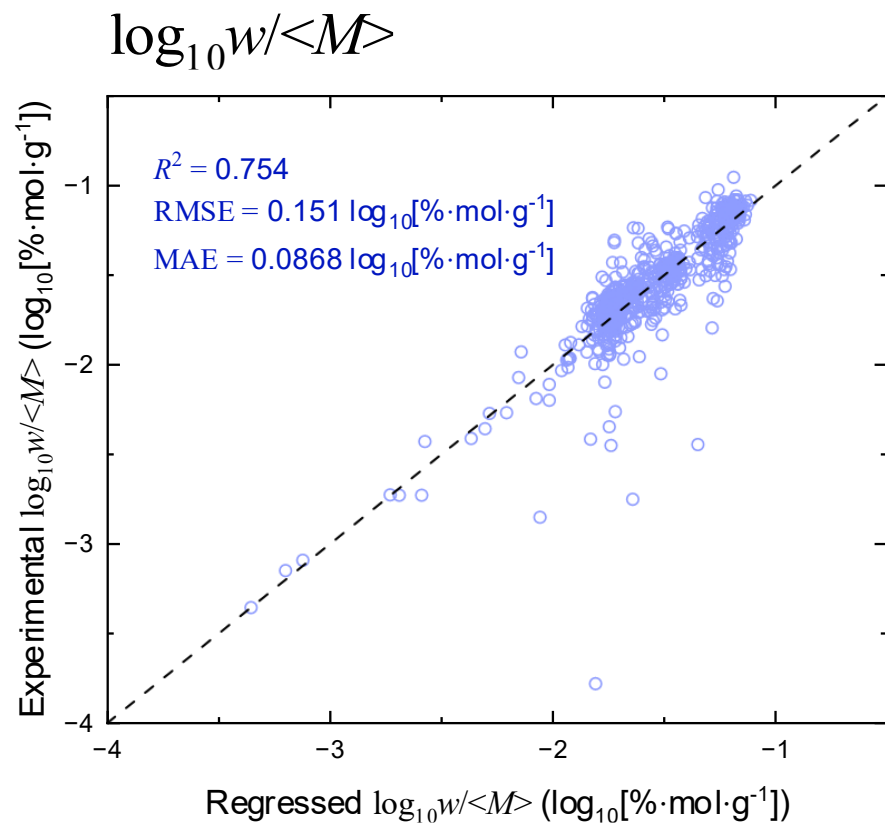
**Task: Given 57 features (candidate descriptors),
take high(est)- R^2 models among 1.95×10^{463} possible formulae!**

Why symbolic regression? Accuracy plus interpretability

- **5-fold test:** Benchmarking used identical data splits across *GoodRegressor* and conventional ML baselines.
- *GoodRegressor* remains competitive with black-box methods such as RandomForest, XGBoost and LightGBM.
- Unlike black-box models, each model is an ensemble set of 10 **explicit equations wherein “important” descriptors can be probed** by using a stacking ensemble of symbolic models, then select descriptors that appear frequently with large standardized coefficients.



Final stacking-ensembled symbolic models



Excellent R^2 ! White-box!

Notes. Ten individual symbolic models were generated for each target and combined by stacking. The models are trained on the curated interstitial hydride dataset without a held-out test set for the final descriptor analysis.

One exemplary equation for each target metric

Supplementary Table 3 An example of symbolic model for $\log_{10}(w/\langle M \rangle)$. Here, $\chi_{\Delta H} = \chi - \chi_H$.



Term index i	Coefficient c_i	Z-scored coefficient $z(c_i)$	Term t_i
1	8.65×10^{-1}	0.498	$\langle \Omega \rangle R_{Tr}$
2	-1.17	-0.409	$\cos\left(\frac{\pi}{200} \frac{\langle V_p \rangle}{R_{Tr+RE}}\right)$
3	-1.92×10^{-1}	-0.225	$10^{\frac{\langle \theta \rangle}{\langle \theta_D \rangle}}$
4	-8.93	-0.221	$\sin\left(\frac{\pi}{10} \langle \chi_{\Delta H} \rangle \langle r_M \rangle\right) \left(\text{erf}\left(\langle v \rangle R_{Tr(TV)} - \frac{1}{20}\right)\right)^{-\frac{1}{2}}$
5	1.40	0.198	$\left(\text{erf}\left(100 \frac{\langle \eta_{FM} \rangle}{\langle V_p \rangle}\right)\right)^{-\frac{1}{2}}$
6	9.71×10^{-1}	0.170	$\cos\left(\frac{\pi}{200} \langle \kappa \rangle \langle \Omega_G \rangle\right)$
7	3.84×10^{-16}	0.161	$\frac{\langle \Omega \rangle \langle V_p \rangle \langle V_p \rangle}{\left(10^{\sin\left(\frac{\pi}{1000} \frac{\langle r_M \rangle}{R_{Tr}}\right) \text{erf}\left(\langle \sigma \rangle \langle \alpha \rangle - \frac{1}{20}\right)}\right)^{\frac{1}{2}}} \times \frac{\langle \theta_D \rangle R_{Tr}}{\langle \alpha \rangle \langle \rho_{mol} \rangle \langle \kappa \rangle \exp(\sigma \langle \chi_{\Delta H} \rangle)} \times \frac{\langle G \rangle \langle \kappa \rangle}{\left(\log_{10}\left(\frac{\langle n_{ve} \rangle}{\langle G \rangle}\right)\right)^{-\frac{1}{2}}} \left(\frac{\langle \rho_{mol} \rangle}{\langle G \rangle \langle \kappa \rangle \langle \rho \rangle \left(\log_{10}\left(\frac{\langle n_{ve} \rangle}{\langle G \rangle}\right)\right)^{-\frac{1}{2}}}\right)^{-\frac{1}{2}} \left(\exp(r \langle \chi_m \rangle)\right)^{-\frac{1}{2}}$
8	4.29×10^{-6}	0.121	$\langle B \rangle \langle \theta_D \rangle$
9	2.08×10^{-1}	0.117	$\sin\left(\frac{\pi}{10} \langle \chi_{\Delta H} \rangle \langle r_M \rangle\right) \sigma(\rho)$
10	-3.27×10^{-1}	-0.111	$\left(10^{\left(\text{erf}\left(\langle v \rangle R_{Tr(TV)} - \frac{1}{20}\right)\right)^{-\frac{1}{2}} \left(\text{erf}\left(100 \left(\log_{10}\left(\frac{\langle \rho \rangle \langle v \rangle}{\langle G \rangle}\right)\right)^{-\frac{1}{2}} \sin\left(\frac{\pi}{1000} \sigma \langle \theta_D \rangle\right)\right)^{-\frac{1}{2}}}\right)^{-\frac{1}{2}}$
11	-2.43×10^{-6}	-0.0978	$\left(\log_{10}\left(\frac{\langle M \rangle}{R_{Tr}}\right)\right)^{-\frac{1}{2}} \langle n_{ve} \rangle \langle \theta_D \rangle \cos\left(\frac{1}{1000} \langle M \rangle \langle \theta_D \rangle\right)$
12	4.29×10^{-2}	0.0793	$\cos(\pi \langle \chi_{\Delta H} \rangle \langle \rho \rangle)$

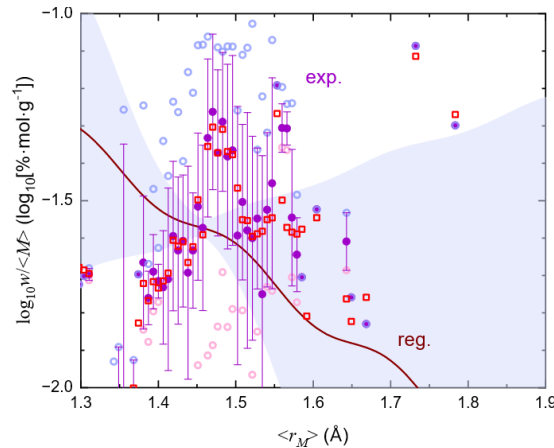
Supplementary Table 4 An example of symbolic model for $\log_{10} P_{eq,RT}$. Here, $\chi_{\Delta H} = \chi - \chi_H$.

Term index i	Coefficient c_i	Z-scored coefficient $z(c_i)$	Term t_i
1	5.13×10^{-1}	3.841	$\langle G \rangle R_{Tr+RE}$
2	1.19×10^9	1.929	$\sin\left(\frac{\pi}{20} \frac{\langle Z \rangle \langle \alpha \rangle}{\langle r_M \rangle \langle \theta_D \rangle}\right)$
3	-8.68×10^{-5}	-1.778	$\frac{\langle G \rangle \langle \theta_D \rangle}{\langle v \rangle R_{Tr+RE}}$
4	1.76×10^2	1.490	$\frac{\langle \rho_{mol} \rangle}{R_{Tr+RE}}$
5	7.22×10^{10}	1.328	$\left(\exp\left(\frac{\langle \alpha \rangle}{\langle \theta_D \rangle} \frac{\langle M \rangle}{\langle \rho_{mol} \rangle}\right)\right)^{-\frac{1}{2}}$
6	-6.18×10^1	-1.001	$\sin\left(\frac{\pi}{2000} \langle B \rangle\right)$
7	-9.58×10^3	-0.898	$\langle G \rangle \langle \alpha \rangle$
8	-2.39	-0.779	$(\log_{10}(\langle M \rangle \langle Z \rangle))^{-\frac{1}{2}}$
9	-3.85×10^{-2}	-0.750	$\sigma(\theta_D)$
10	1.33×10^{-6}	0.705	$\langle n_{ve} \rangle \langle B \rangle \frac{\langle \kappa \rangle}{\langle \Omega_G \rangle}$
11	9.78×10^{-4}	0.468	$\langle \chi_{\Delta H} \rangle \langle M \rangle \frac{\langle r_M \rangle}{\langle \Omega_G \rangle}$
12	1.57	0.456	$\sin\left(\frac{\pi}{20} \text{erf}\left(\frac{1}{10} r(B)\right) \langle \chi_{\Delta H} \rangle \langle M \rangle\right)$
13	-3.92×10^{-2}	-0.379	$\langle G \rangle R_{Tr(TV)}$
14	-1.17	-0.288	$\sigma(Z)$
15	9.06×10^{-2}	0.213	$\frac{\langle Z \rangle}{\langle r_M \rangle} r(\chi_{\Delta H})$
16	-1.09×10^{-2}	-0.204	$\langle \chi_{\Delta H} \rangle \langle M \rangle \sigma(\rho)$
17	-4.75×10^{-3}	-0.204	$\langle \kappa \rangle r(n_{ve})$
18	6.72×10^{-2}	0.194	$(\sigma(n_c))^{-\frac{1}{2}}$
19	1.18×10^{-1}	0.0956	$r(\kappa)$
Intercept	-7.22×10^{10}		

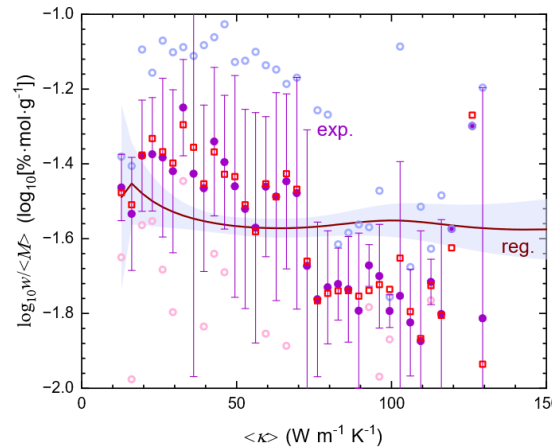
Capacity descriptors: radius sweet spot and softer host lattice

Average metal radius Average thermal conductivity

$\langle r_M \rangle$



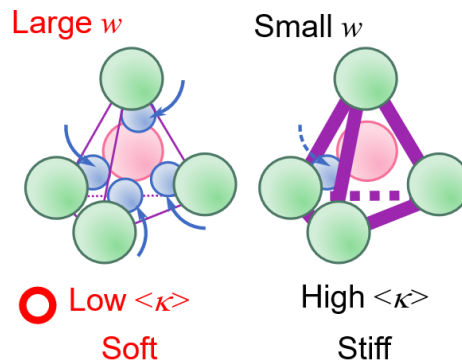
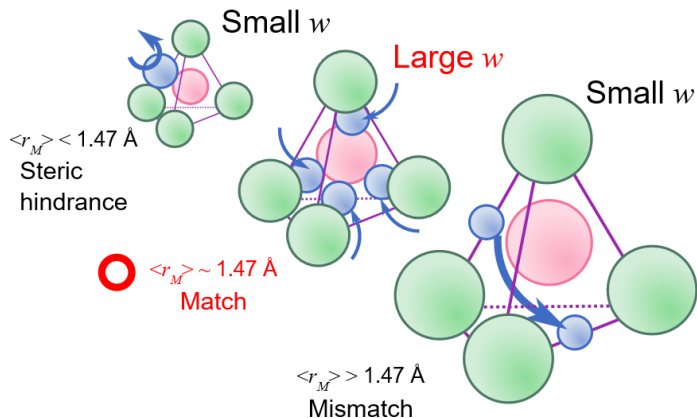
$\langle \kappa \rangle$



- $\langle r_M \rangle$ is the strongest descriptor: optimum around 1.47 Å.

- Too small: steric constraints; too large: cage/H mismatch.

- Lower $\langle \kappa \rangle$ tends to support higher capacity, consistent with softer/more adaptable lattices.

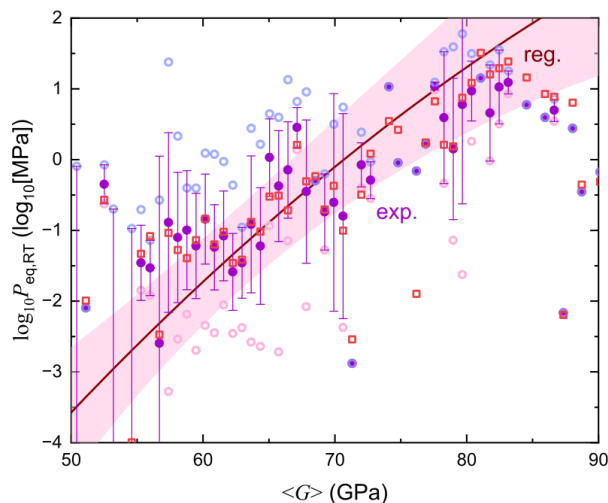


Design rule for w : tune $\langle r_M \rangle$ near 1.47 Å and avoid very stiff/high- κ hosts.

Equilibrium pressure descriptors: elastic response controls stability

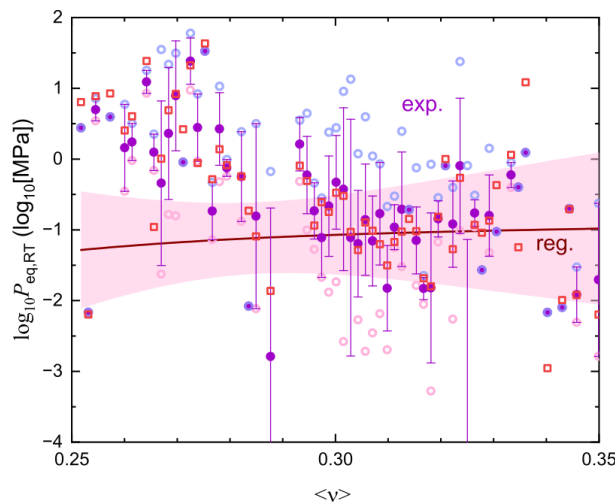
Average shear modulus

$\langle G \rangle$

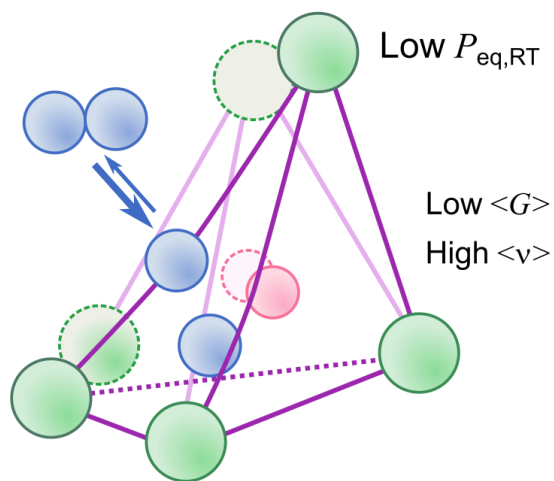


Average Poisson ratio

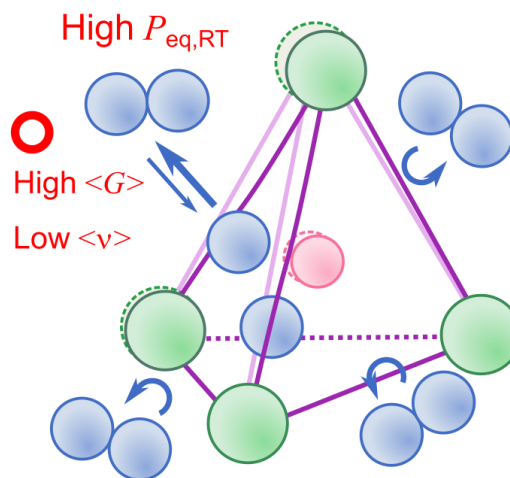
$\langle \nu \rangle$



- $\langle G \rangle$ is the dominant descriptor for $P_{eq,RT}$.
- Higher shear modulus increases elastic penalty for H insertion, destabilizing hydride formation.
- Poisson ratio ν captures mechanical compliance; behavior changes around $\nu \approx 0.3$.
- Pressure is therefore governed by host-lattice rigidity/deformability.



Deformable (Metal hydride is stable)



Robust (Metal hydride is unstable)

Design rule for $P_{eq,RT}$: use elastic descriptors to keep hydrides neither too stable nor too unstable.

Materials design: optimize compositions under practical pressure

Optimization objective

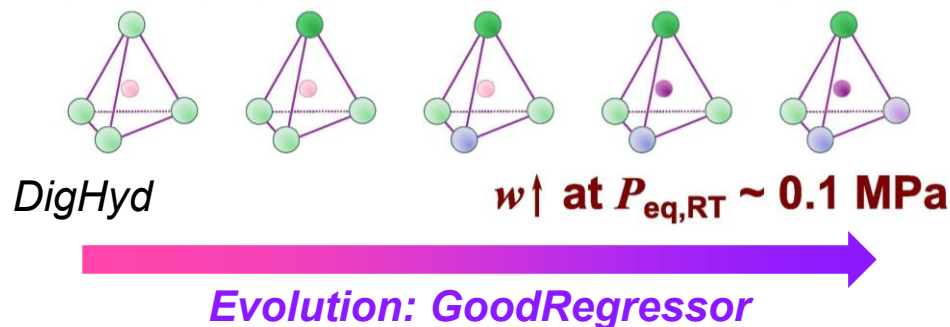
maximize predicted w

subject to $-1.5 \log_{10}[\text{MPa}] < \log_{10} P_{\text{eq,RT}}$
 $< -0.5 \log_{10}[\text{MPa}]$

Search operations

- Start from experimental *DigHyd* compositions with multi-temperature PCT data.
- Preserve stoichiometry and parent structure class; Allow full or 10% metal substitutions.
- Restrict substitutions by electronegativity ($\Delta < 0.5$) and metallic-radius differences ($\Delta < 0.5 \text{ \AA}$).

Physically-grounded materials design



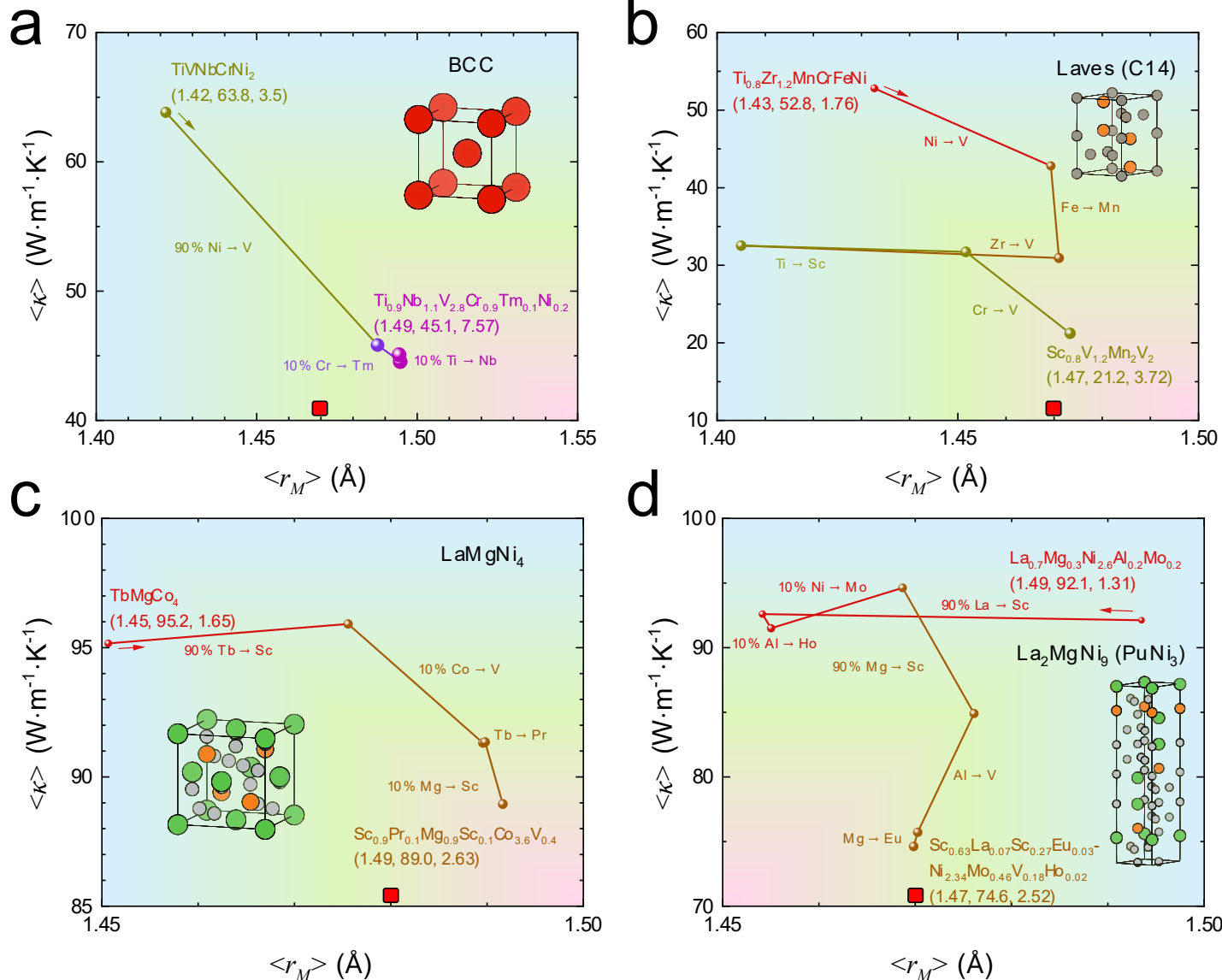
Reliability filter:

retain low-ensemble-uncertainty candidates.

For materials optimization, the *Designer* module described in ref. 23 was employed, using the *DigHyd* dataset as input. During the search for optimization pathways from existing experimental compositions, only candidate compositions yielding ensemble standard deviations below $0.5 \log_{10}[\% \text{ mol g}^{-1}]$ and $0.7 \log_{10}[\text{MPa}]$ for $\log_{10}(w/M)$ and $\log_{10} P_{\text{eq,RT}}$, respectively, were considered. For the final optimized compositions, a stricter criterion was applied: only those with ensemble standard deviations below $0.1 \log_{10}[\% \text{ mol g}^{-1}]$ were retained for $\log_{10}(w/M)$.

The output is a physically grounded pathway, not a claim of immediate synthesizability.

Optimization pathways: descriptor-guided movement



- Representative best candidates are selected per parent structure class.

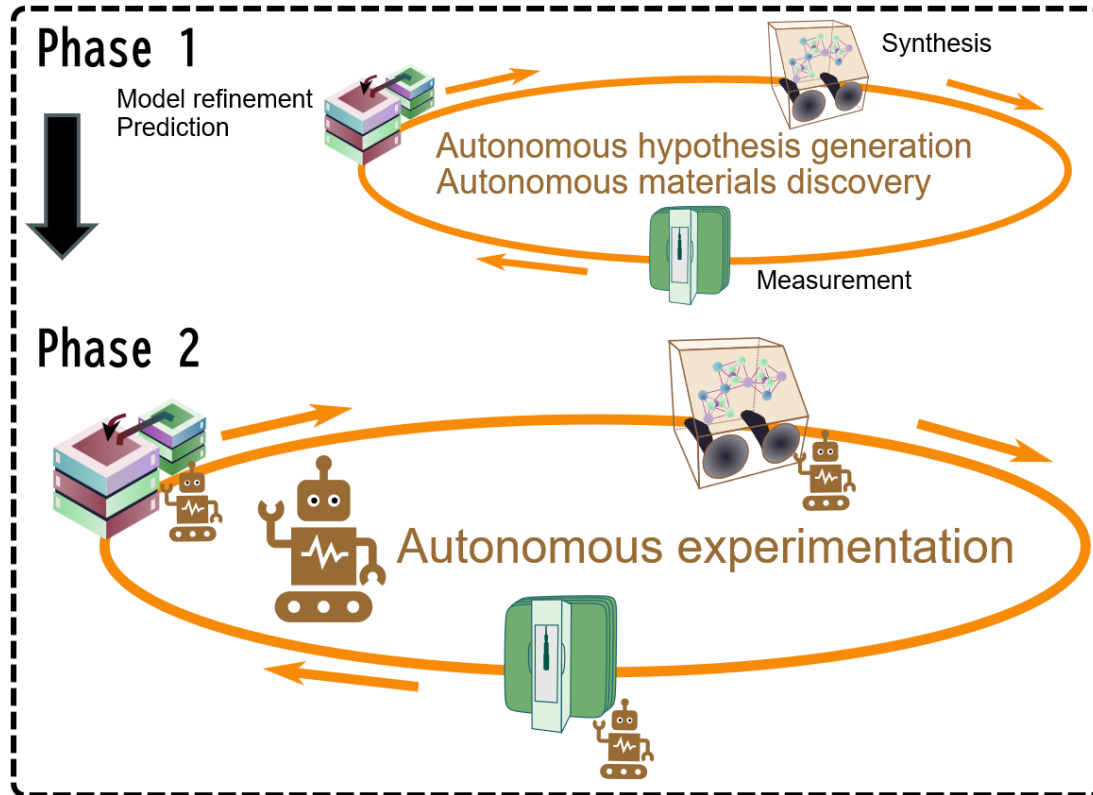
- The trajectories self-consistently update descriptors with composition.

- Most paths move toward $\langle r_M \rangle \approx 1.47$ Å, while $\langle \kappa \rangle$ decreases or stays nearly unchanged, given the optimal point (the red rectangles).

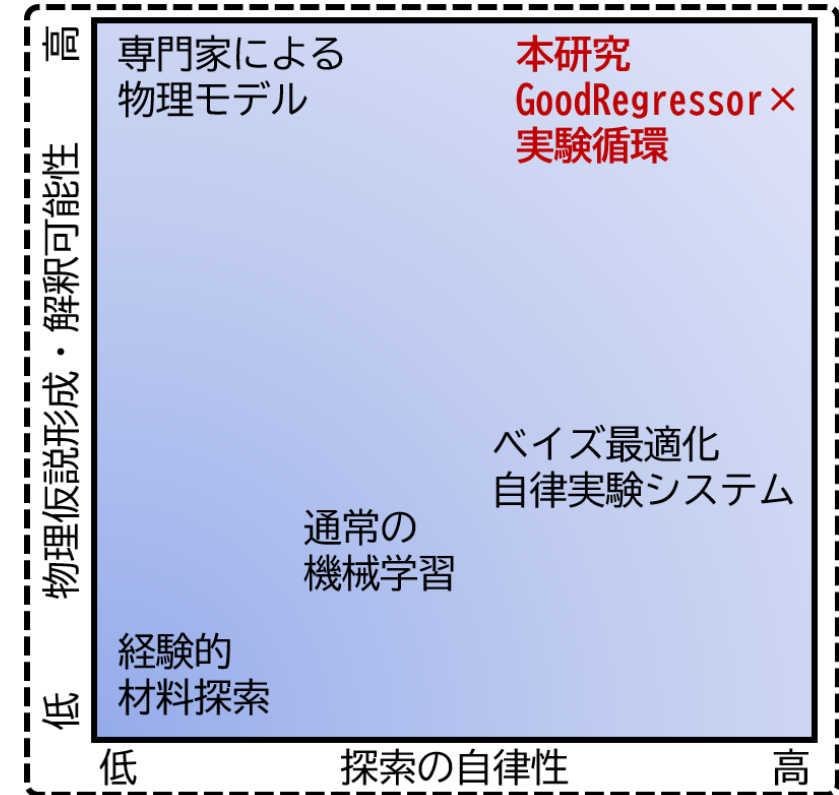
Common direction: match cage size + soften lattice.

Highest-capacity four of the eight representative pathways shown.

Symbolic regression: auspicious sign of automatic discovery?



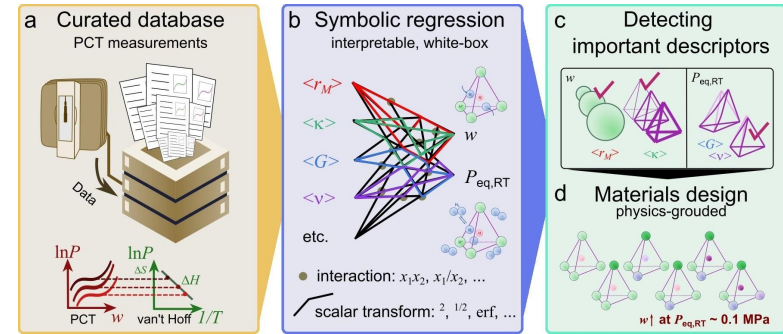
Don't we need to confirm which ML would be effective before jumping into "autonomous experimentation"? Don't we need a **test bed**?



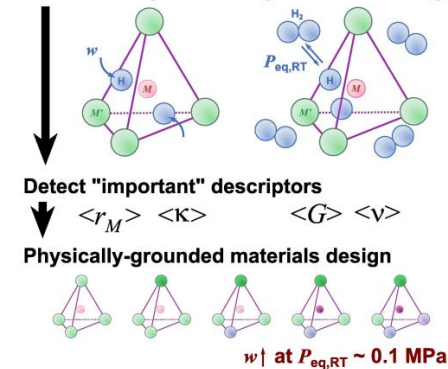
Symbolic regression modeling is **aware of physics, which humans can directly see!** The observed physics (or knowledge) can be **updated** by adding datapoints.

Conclusion and perspectives

- Curated single-/near-single-phase PCT data enable robust descriptor learning for interstitial hydrides.
- Capacity w is governed mainly by cage-size matching ($\langle r_M \rangle \approx 1.47 \text{ \AA}$) and lattice adaptability ($\langle k \rangle$).
- Equilibrium pressure $P_{\text{eq,RT}}$ is governed mainly by elastic response ($\langle G \rangle$ and $\langle \nu \rangle$).
- Optimization pathways use these separated mechanisms to propose higher- w candidates near practical pressure.



White-box "symbolic" regression modeling



Next step: synthesize and measure the candidate pathways, then close the loop with updated *DigHyd* data.

Thank you